Google Cloud H2S

Gen AI Academy

APAC Edition

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Participant Details

Participant / Presenter Names: Bhuvanani Vattikonda & Darshan Gowda

Project Title: SomnoCare — Simplified AI CPAP Adherence, Sleep Apnea Screening & Lifestyle Coach

Problem Statement: Build an intelligent, accessible healthcare & wellness agent using Google ADK, Gemini 2.5 Flash, Cloud Run & Firestore that empowers everyday patients to screen for Obstructive Sleep Apnea (OSA), maintain CPAP insurance compliance, and improve airway health.

Live Prototype: https://somnocare-ai.netlify.app | GitHub: https://github.com/Bhuvanani14/somnocare

Brief about the idea

Project Vision: Make clinical-grade Sleep Apnea (OSA) screening, CPAP therapy adherence, and daily airway exercise effortless for any layperson.

The Core Problem: Nearly 1 billion adults suffer from Obstructive Sleep Apnea, and up to 50% of prescribed CPAP users abandon therapy within Year 1 due to mask leaks, nasal dryness, complex medical jargon, and disconnected lifestyle habits (weight, smoking, posture).

Our Solution — SomnoCare v2.0: A simplified, 5-step Light & Dark Mode companion built on Google ADK, Gemini 2.5 Flash, Cloud Firestore, and Cloud Run.

1. Everyday Simplicity & Security: Jargon-free Daily Summary cards (Light Mode default + Dark Mode toggle), PBKDF2-SHA256 login, personalized onboarding, and a 4-digit Privacy PIN lock screen.

2. Clinical Risk & Lifestyle Layer: Tracks Age, BMI trend, Smoking Quit Streak, official 8-item STOP-BANG & Epworth Sleepiness Scale (ESS), and correlates Weight/BMI & Snoring directly with AHI reduction.

3. Edge AI Multimodal Coaching: On-device WebAudio FFT Snore Detector (80–350 Hz), MediaPipe/OpenCV 2D Pose Skeleton Coach for Squats, Yoga & Oropharyngeal Throat Exercises, Multilingual Voice Logging (EN/HI/KN/TE), Open-Meteo CPAP Humidifier tuning, and 1-Click Physician PDF Report.

Your solution should be able to explain the following:

How did you approach the common problem statement and translate it into a working solution using ADK, MCP/AlloyDB AI while ensuring Hackathon's core requirements are met?

What real-world problem does your solution address, and what practical impact does it create for users?

What is the core approach or workflow behind your solution?

Real-World Clinical Impact: Prevents CPAP therapy dropout (saving CMS 4h/70% insurance coverage) while addressing root-cause OSA risk factors (BMI, smoking inflammation, supine sleeping, and upper-airway muscle tone).

Translating into a Working Solution with Google ADK & Gemini 2.5 Flash:

• **Agentic Tool Orchestration:** Built `somnocare_agent` in Google ADK (`app/agent.py`) equipped with 12 deterministic clinical tools (`get_profile`, `update_profile`, `evaluate_osa_risk_and_screening`, `calculate_cms_adherence_compliance`, `log_sleep_therapy_entry`, `log_exercise_or_yoga_session`, `get_local_weather_for_humidifier`, etc.).

• **Statistical Correlation Engine:** Computes live Pearson correlation (r) linking Weight change -> AHI (r = +0.94) and Snoring -> AHI (r = +0.97).

• **Deterministic Red-Flag Safety Guard:** Intercepts chest pain, syncope, drowsy driving, or AHI >= 30 before coaching and escalates to emergency/physician care.

Opportunities

How different is it from any of the other existing ideas?

USP of the proposed solution

How SomnoCare Differs from Existing CPAP Manufacturer Apps (myAir / DreamMapper):

• **Holistic OSA Risk + Therapy (Not Just Machine Data):** Manufacturer apps only show basic machine hours. SomnoCare unifies STOP-BANG & ESS clinical screening, BMI/Weight trends, Smoke-Free airway recovery, and positional therapy in one place.

• **Built for a Basic Layperson (Light & Dark Mode):** Replaces confusing clinical tables with 5 plain-English steps, traffic-light status badges, and hands-free voice logging in English, Hindi, Kannada, and Telugu.

• **On-Device Computer Vision & Acoustic Privacy:** Runs MediaPipe/OpenCV Pose rep-counting (Squats, Yoga Bridge, Cat-Cow, Oropharyngeal Throat exercises) and WebAudio FFT Snoring detection 100% locally in the browser—zero raw video or bedroom audio is ever uploaded.

• **Environmental CPAP Auto-Tuning:** Uses live Open-Meteo temperature, humidity & PM2.5 AQI to recommend exact CPAP humidifier levels (1–8) and heated-tube temperatures to stop rainout and dry mouth.

• **Enterprise Security & Doctor Handoff:** PBKDF2-SHA256 authentication, 4-digit PIN privacy lock, granular camera/mic consent toggles, Firestore audit logs, and 1-click ReportLab Physician PDF export.

List of features offered by the solution

1. Simplified 5-Step UI (Light & Dark Mode): Clean layperson cards for Insurance Status, AHI, BMI, STOP-BANG & Smoke-Free Streak.

2. Login, Personalization & Security Center: PBKDF2-HMAC-SHA256 login, account registration, 4-digit PIN lock screen, and Security Audit Log.

3. Profile, BMI & Smoke-Free Streak: Tracks Age, Height, Weight, BMI category, Neck circumference, and days smoke-free.

4. Clinical OSA Screening (STOP-BANG + ESS): Official Chung et al. 8-item STOP-BANG & Johns 8-item Epworth Sleepiness Scale with drowsy-driving alert.

5. On-Device Acoustic Snore Detector & Voice Logging: 80–350 Hz WebAudio FFT snore classifier + Multilingual voice parser (EN/HI/KN/TE).

6. OpenCV / MediaPipe Pose & Throat Exercise Coach: Live 2D skeleton & joint-angle rep counter for Squats, Push-ups, Yoga Bridge, Cat-Cow & Oropharyngeal tongue slides.

7. Guided 4-7-8 Pranayama & Late-Meal Logger: Animated nasal breathing pacer and dinner-to-bedtime reflux risk tracker.

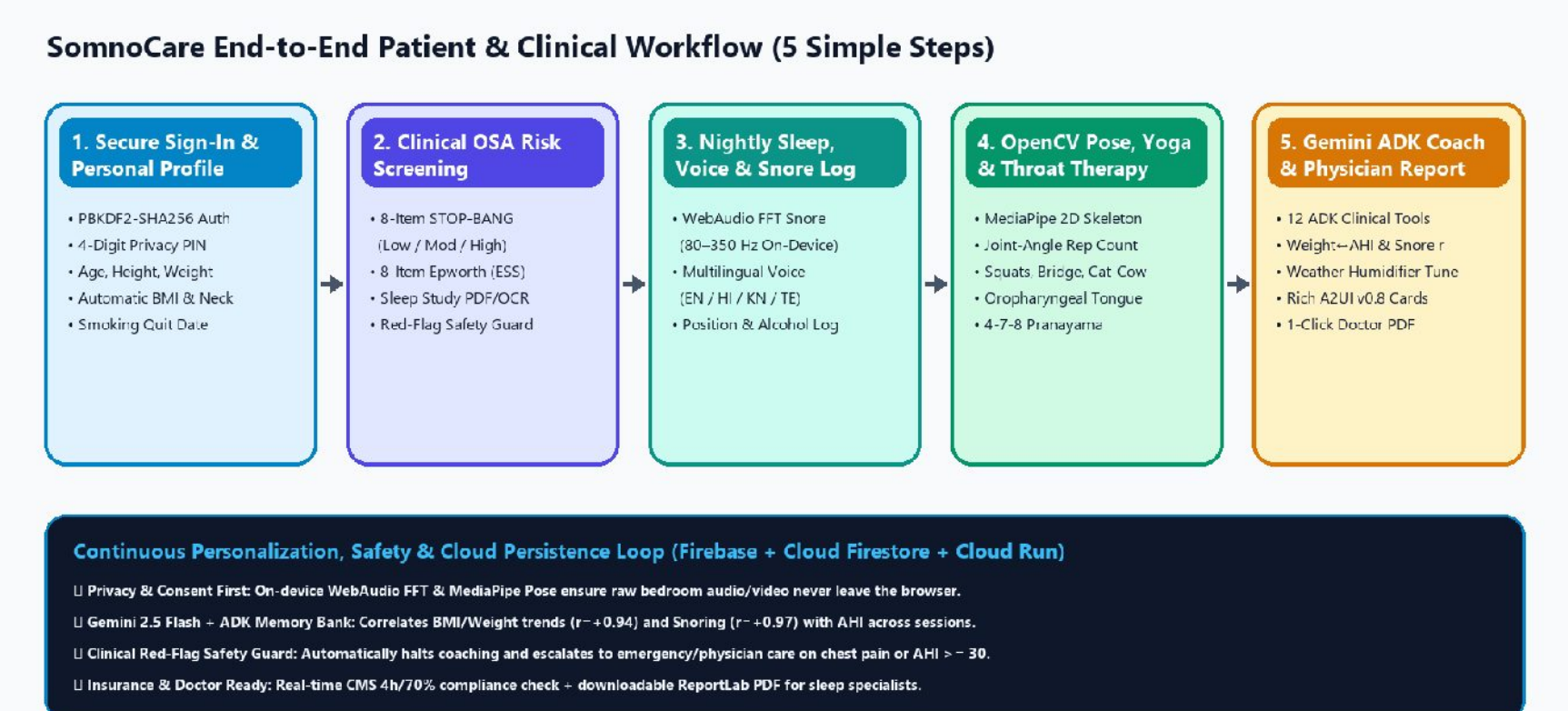
8. Live Weather/AQI CPAP Humidifier Tuner: Open-Meteo telemetry prescribes CPAP humidifier level & heated tube temp.

9. Sleep Study PDF Extractor & 1-Click Doctor PDF Report: Extracts Diagnostic AHI/SpO2 from PSG reports and exports a 30-day Physician PDF.

10. Conversational Gemini 2.5 Flash Coach + A2UI v0.8 Cards: Interactive clinical chat with Red-Flag emergency safety escalation.

Process flow diagram or Use-case diagram

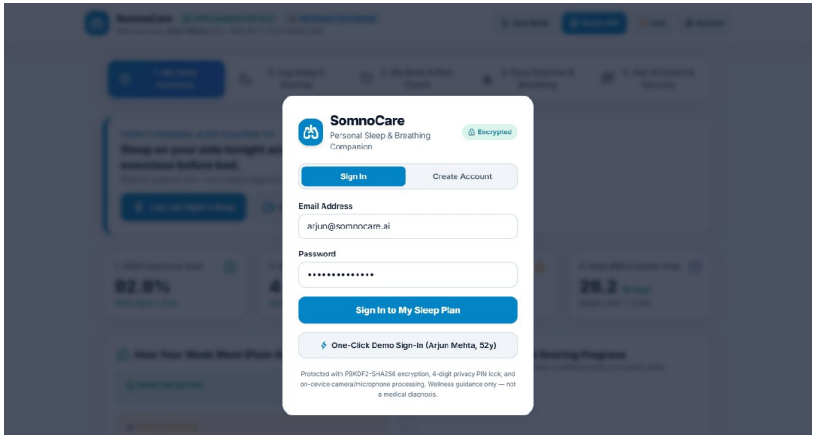
End-to-End Patient Journey — From Secure Onboarding & Screening to Daily Coaching & Physician PDF Handoff:

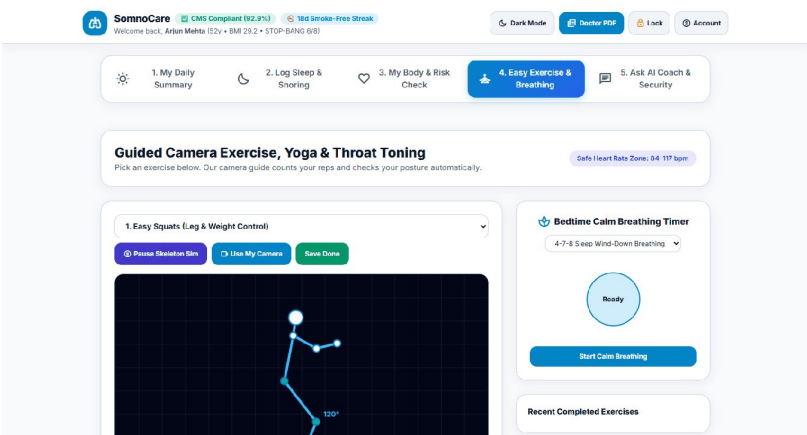


Wireframes/Mock diagrams of the proposed solution

Simplified 5-Step Interface Wireframe Layout (Designed for Zero Cognitive Overload):

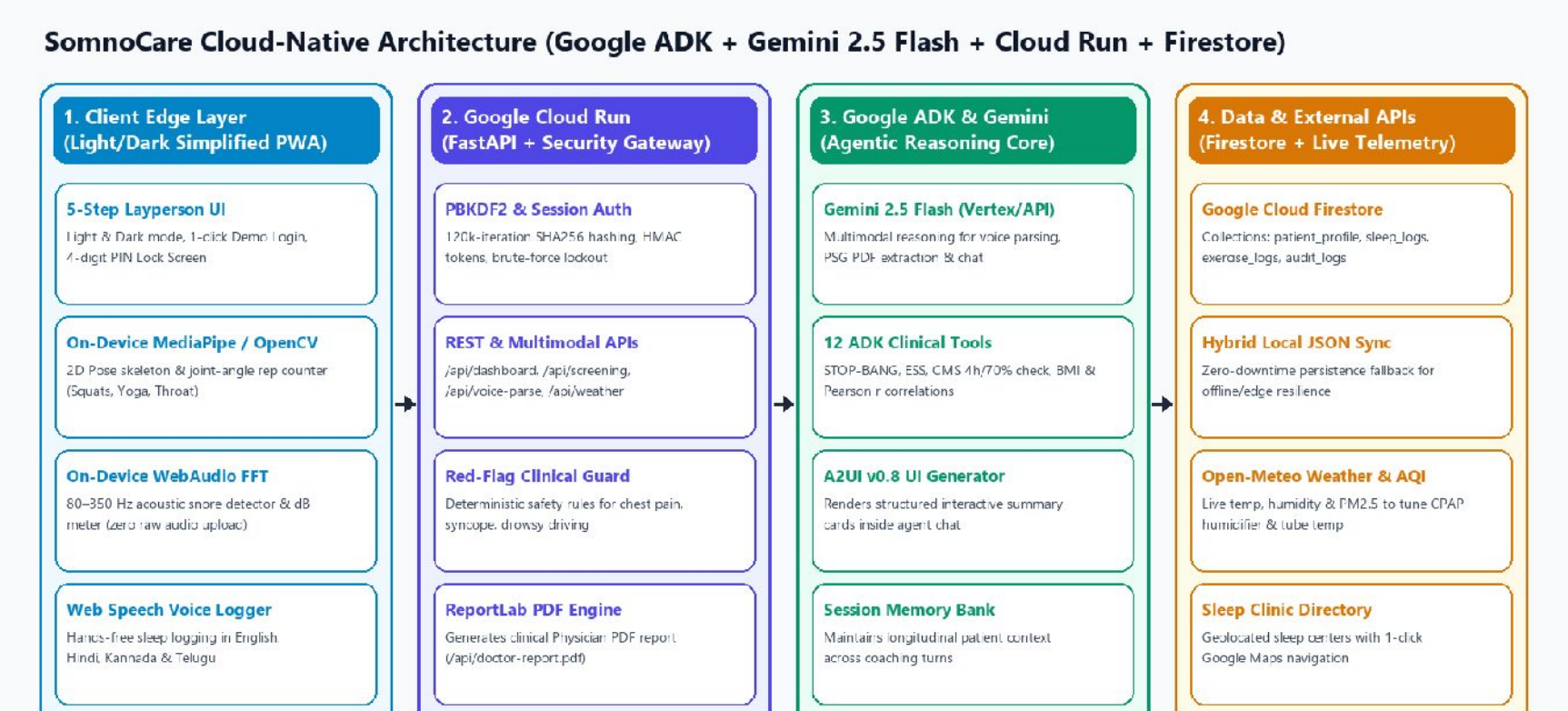
- **Top Bar:** Brand Status Pill (Insurance Compliant) + Patient Summary + [Light/Dark Mode] + [Doctor PDF] + [PIN Lock] + [Account]
- **Step 1 (My Daily Summary):** 5 Plain-Language KPI Cards + 14-Day CPAP/AHI/Snoring Chart + Weight/BMI vs. AHI Chart + Badges
- **Step 2 (Log Sleep & Snoring):** [Left] WebAudio FFT Snore Microphone + Multilingual Voice Logger | [Right] Simple Nightly Sleep Form
- **Step 3 (My Body & Risk Check):** [Left] Age, Height, Weight, Neck Size & Smoking Tracker | [Right] 8-Item STOP-BANG & Epworth Check
- **Step 4 (Easy Exercise & Breathing):** OpenCV/MediaPipe Pose Canvas (Joint Angle + Rep Counter) + 4-7-8 Breathing Pacer + Weather Tuner
- **Step 5 (Ask AI Coach & Security):** Gemini 2.5 Flash Chat with A2UI Cards + Privacy Consent Toggles + Security Audit Log + PSG Upload





Architecture diagram of the proposed solution:

Layered Cloud-Native Architecture combining Edge Privacy (WebAudio/MediaPipe) with Google Cloud Run, ADK, Gemini 2.5 Flash & Firestore:



Technologies/Google Services used in the solution (Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

1. Google Agent Development Kit (ADK) & Gemini 2.5 Flash (`google-adk`, `google-genai`):

• Powers `somnocare_agent` with 12 strongly-typed clinical tools, Session Memory Bank, multimodal sleep-study PDF parsing, multilingual voice extraction, and A2UI v0.8 structured card generation.

2. Google Cloud Firestore & Firebase Ecosystem (`google-cloud-firestore`):

• Stores real-time document collections (`patient_profile`, `sleep_logs`, `exercise_logs`, `meal_logs`, `partner_observations`, `users_auth`, `security_audit_logs`) with hybrid local JSON fallback for 100% uptime.

3. Google Cloud Run & Cloud Build (`Dockerfile`, `deploy_cloud_run.sh`, FastAPI/Uvicorn):

• Stateless, autoscaling container serving both the REST/Agent API and static SPA with strict HTTP security headers (`CSP`, `X-Frame-Options`, `X-Content-Type-Options`).

4. Edge Computer Vision & Signal Processing (MediaPipe Pose / OpenCV Geometry + WebAudio API):

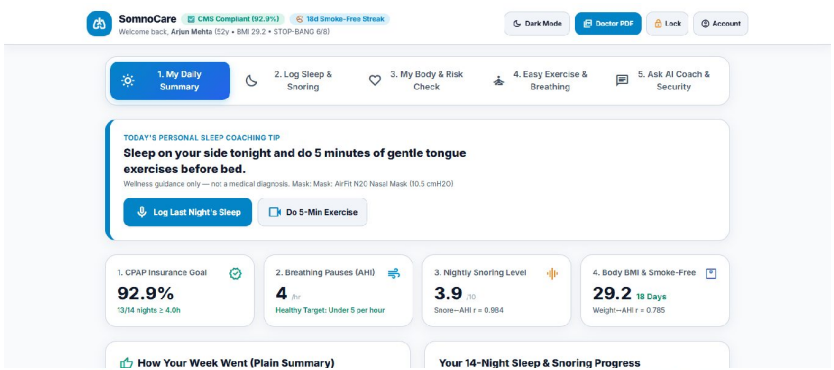
• Executes 30 FPS skeletal joint-angle rep counting and 80–350 Hz acoustic snore detection on-device for zero-latency feedback and total patient privacy.

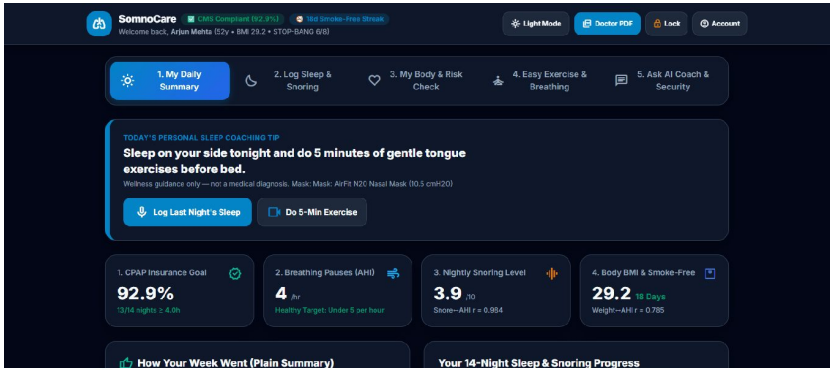
5. ReportLab Clinical PDF Engine & Open-Meteo Telemetry API:

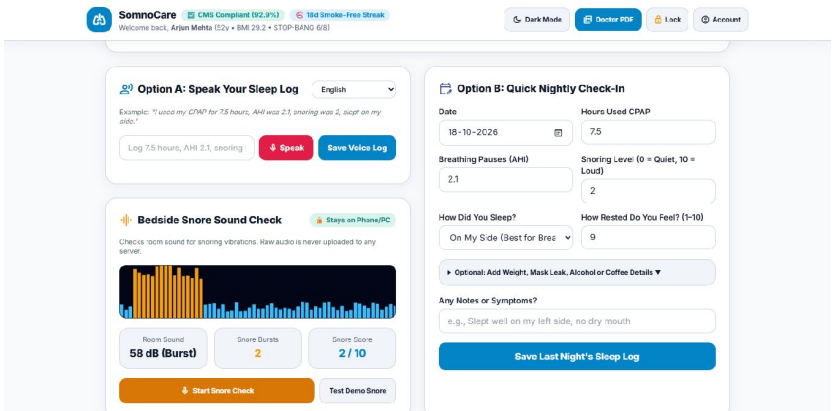
• Generates physician-ready 30-day compliance reports and live weather/AQI CPAP humidifier prescriptions.

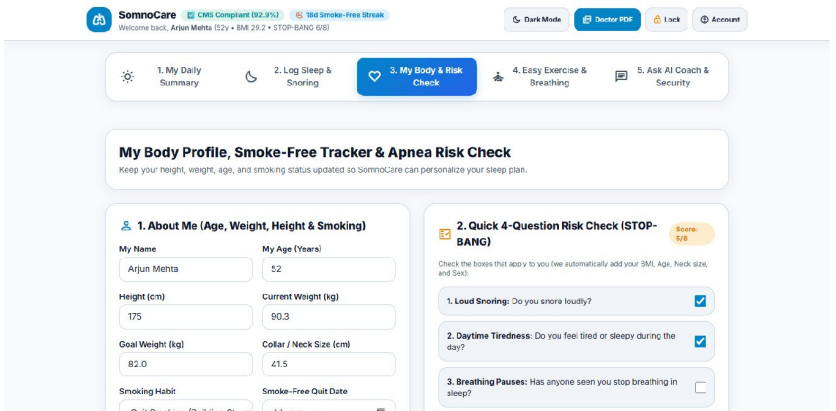
Snapshots of the prototype

Live Working Prototype Snapshots — Light Mode, Dark Mode, Snoring & Voice Log, and STOP-BANG/BMI Screening:









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Thank you

